

2016.0 RANGE ROVER (LG), 303-05

ACCESSORY DRIVE – TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the accessory drive system and operation, refer to the relevant Description and Operation section of the workshop manual.

REFER to: [Accessory Drive](#) (303-05A Accessory Drive - TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.

2. Visually inspect for obvious signs of mechanical damage.

Visual Inspection

MECHANICAL
▪ Auxiliary drive belt condition (cracking/damage/contamination) ▪ Idler assembly ▪ Generator ▪ Engine cooling fan ▪ Tensioner assembly ▪ Engine coolant pump ▪ Air Conditioning (A/C) compressor ▪ Torsional vibration damper ▪ Dynamic response pump ▪ Tensioner assembly ▪ Auxiliary drive belt ▪ Security/correct installation of the fuel injection pump cover ▪ Fuel injection pump belt condition (cracking/damage/contamination) ▪ Fuel injection pump belt tensioner assembly ▪ Fuel injection pump ▪ Fuel injection pump belt

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

CAUTION:

If the engine is run without the auxiliary drive belts connected to eliminate driven components, diagnostic trouble codes, (DTCs) may be set which must be cleared before the vehicle is returned to the owner. The engine should not be run for more than 2-3 minutes with the belts disconnected. Failure to follow this instruction may result in damage to the vehicle.

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART (ACCESSORY DRIVE BELT)

SYMPTOM	POSSIBLE CAUSES	ACTION
Noise	▪ Auxiliary drive belt condition ▪ Auxiliary drive belt tension ▪ Pulleys misaligned ▪ Driven components (including tensioners)	▪ Check the auxiliary drive belt condition (see visual inspection) ▪ Check the auxiliary drive belt tensioner function ▪ Check the pulley alignment ▪ Check the driven components for excessive resistance to rotation ▪ Using the manufacturer approved diagnostic system, perform routine - Noise, vibration and harshness sensor diagnostic test - Accessory drive belt
Auxiliary drive belt does not hold tension	▪ Belt condition ▪ Tensioner fault	▪ Check the auxiliary drive belt condition (see visual inspection) ▪ Check the auxiliary drive belt tensioner function

SYMPTOM CHART (FUEL INJECTION PUMP BELT)

SYMPTOM	POSSIBLE CAUSES	ACTION
Noise	- Fuel injection pump belt condition - Fuel injection pump belt fouling cover - Fuel injection pump belt tensioner bearing failure - Fuel injection pump failure	- Check the fuel injection pump belt condition (see visual inspection) - Check the fuel injection pump belt cover for indications of fouling (this may indicate a pump misalignment), refer to the relevant workshop manual section - The fuel injection pump belt tensioner must be renewed if the belt is removed, making a check of the bearing impractical - Remove the fuel injection pump belt, check the fuel injection pump pulley for security. Check the fuel injection pump for excessive resistance to rotation (excessive resistance in the pump will cause the pulley securing nut to loosen as a design feature). Check for diagnostic trouble codes (DTCs) indicating a pump malfunction
Fuel injection pump belt does not hold tension	- Fuel injection pump belt condition - Tensioner fault	- Check the fuel injection pump belt condition (see visual inspection) - Check the tensioner function
Loss of drive (with no drive to the fuel injection pump, the engine will not run)	- Fuel injection pump belt broken / stripped teeth - Drive pulleys	- Investigate the cause of the fuel injection pump belt breakage/damage (a belt broken at a 45 degree angle normally indicates a shear, a break straight across the belt normally indicates that the belt has been crimped). - Check the fuel injection pump for excessive resistance to rotation (excessive resistance in the pump will cause the pulley securing nut to loosen as a design feature). Check for DTCs indicating a pump malfunction

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DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).